SIYI (CELINE) CHEN

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EDUCATION

University of Toronto (St. George Campus) **Toronto, ON**
Honours BSc, Computer Science Specialist, ASIP (Co-op Program) ***Sep 2024- Jun 2028***

- Recipient of the University of Toronto International Scholar Award, valued at up to $100,000
- Relevant Coursework: Software Design, Data Structures and Algorithms, Machine Learning, Computer Organization, Database Systems, Probability and Statistics, Multivariable Calculus, Linear Algebra

SKILLS

Programming Languages: Python, Java, C, SQL
Computer Science: Data Structures and Algorithms, Object-Oriented Programming, Problem Solving, Software Design
Software Engineering: API Integration, Debugging, Testing, Version Control, Clean Architecture, Automation
Machine Learning & Data: Machine Learning, Feature Engineering, Data Analysis, Data Cleaning, Data Visualization
Tools & Technologies: Git, GitHub, Linux/Unix, pandas, NumPy, scikit-learn, Streamlit, Dash, Plotly, Matplotlib

EXPERIENCE

Nomura **Shanghai, China**
Asset Management Data & Operations Intern *May 2025 – Aug 2025*

- Validated fixed-income trade, settlement, NAV, and product datasets used in daily reporting and reconciliation workflows.
- Investigated abnormal values, missing records, and reconciliation discrepancies to improve reporting accuracy and operational reliability.
- Developed Python workflows using pandas and APIs to retrieve, clean, and visualize foreign-exchange and data.
- Automated the generation of charts and structured outputs for repeatable internal analysis and reporting.

PROJECTS

Financial Impact Prediction Model — SDSS Datathon 2026 *Feb 2026*
Tools: Python, pandas, scikit-learn, XGBoost, Streamlit

- Built a multiclass XGBoost model to predict whether an individual's financial condition would improve, decline, or remain stable after an economic shock.
- Cleaned a large financial survey dataset and engineered features involving income, province, debt, deposits, homeownership, and TFSA participation.
- Evaluated model performance using accuracy, macro F1-score, and confusion matrices.
- Deployed the model through a Streamlit application that generates predictions from user-provided inputs.

BikeShare Toronto Trip Planner *Nov 2025*
Tools: Java, OOP, Clean Architecture, Swing, OpenRouteService API, JSON

- Developed a Java desktop application that compares BikeShare and walking routes based on travel time, distance, and cost.
- Designed modular components using Clean Architecture to separate user-interface, business-logic, and data-access responsibilities.
- Integrated the OpenRouteService API and parsed JSON responses to generate route information and travel recommendations.
- Implemented persistent storage to save trip information and support repeated route analysis.

Currency Exchange Rate Tracker *Jul 2025*
Tools: Python, pandas, WindPy API, Matplotlib, Plotly

- Built a Python analytics tool to retrieve, clean, and visualize 20+ years of historical foreign exchange data
- Integrated the WindPy API to process real-time and historical FX market datasets for financial trend analysis
- Implemented interactive filtering and automated chart generation to compare currency movements across custom time periods